

## Citations of Retracted ICPSR Papers

Number of citations:

```
# add file id column to merged - TODO: move to dc/preprocessing file?
merged <- merged |> mutate(file.id = NA, .before = Title)

merged$file.id[which(merged$Title ==
  "Worlds Apart? The Reception of Genetically Modified Foods in Europe and the U.S."
)] = "WorldsApart"
merged$file.id[which(merged$Title ==
  "Nonmetropolitan sex-role ideologies: A longitudinal study"
)] = "NonmetroRole"
merged$file.id[which(merged$Title ==
  "School social bonds, school climate, and school misbehavior: A multilevel analysis"
)] = "SchoolSocialBonds"
merged$file.id[which(merged$Title ==
  "Suicide after natural disasters"
)] = "SuicideNatDisasters"

merged$file.id[which(merged$Title ==
  "Electronic cigarette use and myocardial infarction among adults in the US Population Assessment of T
)] = "ECigUse"
merged$file.id[which(merged$Title ==
  "The verdict is in: How did they decide? Using drivers' self-reported data to understand officers' de
)] = "HowDidTheyDecide"
merged$file.id[which(merged$Title ==
  "Chronic adolescent marijuana use as a risk factor for physical and mental health problems in young a
)] = "MarijuanaUse"
merged$file.id[which(merged$Title ==
  "Tuning in, not turning out: Evaluating the impact of ethnic television on political participation"
)] = "TurningIn"
merged$file.id[which(merged$Title ==
  "Implications of new hypertension guidelines in the United States"
)] = "HypertensionGuidelines"

# add retraction dates to refs
refs <- refs |> mutate(ret.date = NA, .after = year)

for (r in 1:nrow(refs)) {
  ix = which(refs$filepath[r] == merged$file.id)
  refs$ret.date[r] = merged$RetractionDate[ix]
}

# add retraction years to refs
refs <- refs |> mutate(ret.year = NA, .after = ret.date)

for (r in 1:nrow(refs)) {
  ix = which(refs$filepath[r] == merged$file.id)
```

```
refs$ret.year[r] = merged$RetractionYear[ix]
}
```

```
# number of citations per paper
refs |> group_by(filepath) |> count()
```

```
## # A tibble: 9 x 2
## # Groups:   filepath [9]
##   filepath          n
##   <chr>          <int>
## 1 ECigUse          61
## 2 HowDidTheyDecide    4
## 3 HypertensionGuidelines 50
## 4 MarijuanaUse        80
## 5 NonmetroRole        11
## 6 SchoolSocialBonds  267
## 7 SuicideNatDisasters 178
## 8 TurningIn           16
## 9 WorldsApart        465
```

```
# number of citations per paper by year
refsbyyear <- refs |> group_by(filepath, year, ret.year) |>
  summarise(num_citations = n())
```

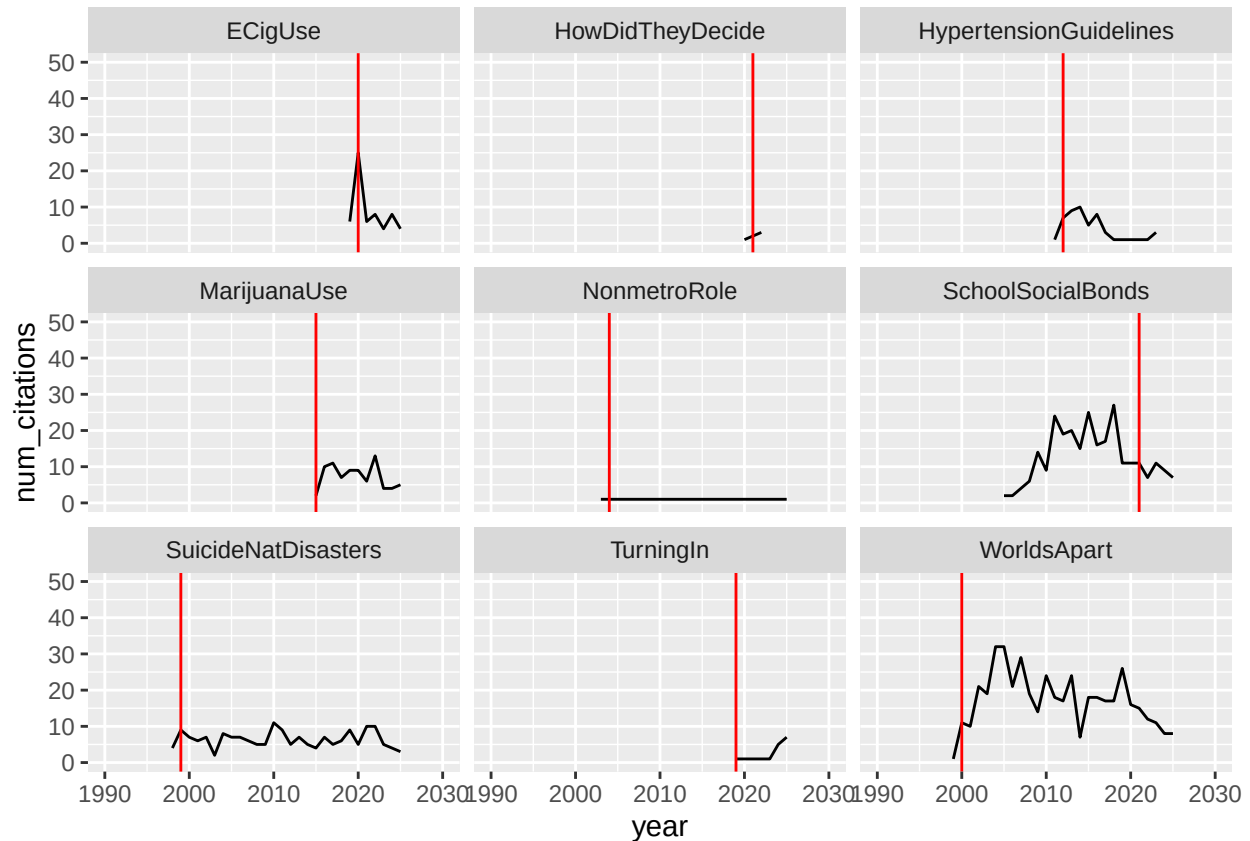
```
## `summarise()` has grouped output by 'filepath', 'year'. You can override using
## the `.groups` argument.
```

```
(head(refsbyyear, 20))
```

```
## # A tibble: 20 x 4
## # Groups:   filepath, year [20]
##   filepath          year ret.year num_citations
##   <chr>          <dbl>   <dbl>         <int>
## 1 ECigUse          2019     2020             6
## 2 ECigUse          2020     2020            25
## 3 ECigUse          2021     2020             6
## 4 ECigUse          2022     2020             8
## 5 ECigUse          2023     2020             4
## 6 ECigUse          2024     2020             8
## 7 ECigUse          2025     2020             4
## 8 HowDidTheyDecide  2020     2021             1
## 9 HowDidTheyDecide  2022     2021             3
## 10 HypertensionGuidelines 2011     2012             1
## 11 HypertensionGuidelines 2012     2012             7
## 12 HypertensionGuidelines 2013     2012             9
## 13 HypertensionGuidelines 2014     2012            10
## 14 HypertensionGuidelines 2015     2012             5
## 15 HypertensionGuidelines 2016     2012             8
## 16 HypertensionGuidelines 2017     2012             3
## 17 HypertensionGuidelines 2018     2012             1
## 18 HypertensionGuidelines 2019     2012             1
## 19 HypertensionGuidelines 2021     2012             1
## 20 HypertensionGuidelines 2022     2012             1
```

Plots of citations by year. Retraction Year in red.

```
# plot citations by year
refsbyyear |>
  group_by(filepath) |>
  ggplot(aes(x = year, y = num_citations)) +
    geom_line() +
    geom_vline(aes(xintercept = ret.year), colour = "red") +
    xlim(1990, 2030) +
    ylim(0, 50) +
    facet_wrap(~filepath)
```



Summary table - number of citations before and after retraction year:

```
refsbyyear |> group_by(filepath, year <= ret.year) |>
  count()
```

```
## # A tibble: 18 x 3
## # Groups:   filepath, year <= ret.year [18]
##   filepath      `year <= ret.year`     n
##   <chr>          <lgl>             <int>
## 1 ECigUse        FALSE                5
## 2 ECigUse        TRUE                 2
## 3 HowDidTheyDecide FALSE                1
## 4 HowDidTheyDecide TRUE                 1
## 5 HypertensionGuidelines FALSE            10
## 6 HypertensionGuidelines TRUE                 2
## 7 MarijuanaUse   FALSE            10
## 8 MarijuanaUse   TRUE                 1
## 9 NonmetroRole   FALSE                 9
```

```
## 10 NonmetroRole      TRUE      2
## 11 SchoolSocialBonds FALSE      4
## 12 SchoolSocialBonds TRUE     17
## 13 SuicideNatDisasters FALSE    26
## 14 SuicideNatDisasters TRUE      2
## 15 TurningIn         FALSE      5
## 16 TurningIn         TRUE       1
## 17 WorldsApart       FALSE    25
## 18 WorldsApart       TRUE       2
```

Flag whether a retracted article was cited the same year it was retracted:

```
# create flag
refs <- refs |> mutate(ref.same.year = (year == ret.year))

# number citing papers retracted during the publication year
refs |> group_by(ref.same.year) |> summarise(n())
```

```
## # A tibble: 2 x 2
##   ref.same.year `n()`
##   <lgl>         <int>
## 1 FALSE         1065
## 2 TRUE           67
```

```
# number with available publication date
refs |> group_by(ref.same.year, !is.na(pub.date)) |> summarise(n())
```

## `summarise()` has grouped output by 'ref.same.year'. You can override using the  
## `.groups` argument.

```
## # A tibble: 4 x 3
## # Groups:   ref.same.year [2]
##   ref.same.year `!is.na(pub.date)` `n()`
##   <lgl>         <lgl>             <int>
## 1 FALSE         FALSE             810
## 2 FALSE         TRUE              255
## 3 TRUE          FALSE              35
## 4 TRUE          TRUE               32
```

Pulling citation data manually for retractions for which `ret.same.year == TRUE`:

```
# setwd(...)

# refs |> filter(ref.same.year == TRUE) |>
#   select(title, authors.list, year, pub.date, doi) |>
#   write.csv(file = "refs_same_year.csv")

# manually pull and add dates to sheet, export to CSV

# Searched on WOS and Scopus. Could typically find data on WOS
# directly, on Scopus, viewed article page on Publisher
# website or on Proquest/SagePub/EBSCOHost through UM library
```

Manually downloaded data:

```
# read edited file into new df
refs_manual <- read.csv("refs_same_year_final.csv")
```

Reformat pub.date

```

month.names <- c(
  'JAN' = 1,
  'FEB' = 2,
  'MAR' = 3,
  'APR' = 4,
  'MAY' = 5,
  'JUN' = 6,
  'JUL' = 7,
  'AUG' = 8,
  'SEP' = 9,
  'OCT' = 10,
  'NOV' = 11,
  'DEC' = 12
)

get.date <- function(datestr) {
  # split into vector of (MONTH, [DAY], YEAR)
  datevec = str_split(datestr, pattern = '\\s') |> unlist()

  # convert year, month to numeric
  # check for date, if yes, convert to numeric
  year = datevec[length(datevec)]
  month = month.names[datevec[1]] |> unname()
  day = ifelse(length(datevec) > 2, datevec[2], NA)

  return(c(year, month, day))
}

refs$pub.year = numeric(nrow(refs))
refs$pub.month = numeric(nrow(refs))
refs$pub.day = numeric(nrow(refs))

for (r in 1:nrow(refs)) {
  # if pub.date data is provided, use get.date function
  if (!is.na(refs$pub.date[r])) {
    datevec = get.date(refs$pub.date[r])
    refs$pub.year[r] = datevec[1]
    refs$pub.month[r] = datevec[2]
    refs$pub.day[r] = datevec[3]
  }
  # if pub.date is not provided, use manually pulled data if available
  else {
    # check if doi is available
    if (!is.na(refs$doi[r])) {
      ix = which(refs$doi[r] == refs_manual$doi)
    }
    # else compare titles, publication years for match
    else {
      ix = which((refs$title[r] == refs_manual$title) &
        (refs$year[r] == refs_manual$year))
    }
  }

  # check whether replacement exists

```

```

if (length(ix) == 0) {
  # if not, insert NAs
  refs$pub.year[r] = NA
  refs$pub.month[r] = NA
  refs$pub.day[r] = NA
}
# if so, replace
else {
  refs$pub.year[r] = refs_manual$pub.year.manual[ix]
  refs$pub.month[r] = refs_manual$pub.month.manual[ix]
  refs$pub.day[r] = refs_manual$pub.date.manual[ix]
}
}
}

```

Flag if month AND year of ref publication are same as retraction month and year

```

# add retraction months to refs
refs <- refs |> mutate(
  ret.month = mdy_hm(ret.date) %>% month(),
  .after = ret.year
)

# add retraction date specifically to refs
refs <- refs |> mutate(
  ret.day = mdy_hm(ret.date) %>% mday(),
  .after = ret.month
)

refs <- refs |> mutate(
  ref.same.month = ((pub.year == ret.year) & (pub.month == ret.month))
)

# number citing papers retracted during the publication month
refs |> group_by(ref.same.month) |> summarise(n())

```

```

## # A tibble: 3 x 2
##   ref.same.month `n()`
##   <lgl>          <int>
## 1 FALSE         313
## 2 TRUE           6
## 3 NA            813

```

More granular before/after retraction counts - only 3 refs published in retraction year for which no month could be obtained (1 publication citing “Suicide after natural disasters...”, 2 citing “Worlds Apart? ...”)

```

# construct pub.after.ret
refs <- refs |> mutate(pub.after.ret = NA)

for (r in 1:nrow(refs)) {
  if (refs$year[r] > refs$ret.year[r]) {
    refs$pub.after.ret[r] = TRUE
  }
  else if (refs$year[r] == refs$ret.year[r]) {
    if (any(is.na(c(refs$pub.month[r], refs$ret.month[r])))) {
      refs$pub.after.ret[r] = NA
    }
  }
}

```

```

    }
    else if (refs$pub.month[r] > refs$ret.month[r]) {
      refs$pub.after.ret[r] = TRUE
    }
    else {
      refs$pub.after.ret[r] = FALSE
    }
  }
  else {
    refs$pub.after.ret[r] = FALSE
  }
}

```

```
refs |> group_by(filepath, pub.after.ret) |> count()
```

```
## # A tibble: 20 x 3
## # Groups:   filepath, pub.after.ret [20]
##   filepath      pub.after.ret      n
##   <chr>          <lgl>         <int>
## 1 ECigUse        FALSE          18
## 2 ECigUse        TRUE           43
## 3 HowDidTheyDecide FALSE           1
## 4 HowDidTheyDecide TRUE            3
## 5 HypertensionGuidelines FALSE           6
## 6 HypertensionGuidelines TRUE           44
## 7 MarijuanaUse   FALSE           1
## 8 MarijuanaUse   TRUE          79
## 9 NonmetroRole   FALSE           2
## 10 NonmetroRole  TRUE            9
## 11 SchoolSocialBonds FALSE          223
## 12 SchoolSocialBonds TRUE           44
## 13 SuicideNatDisasters FALSE           5
## 14 SuicideNatDisasters TRUE          172
## 15 SuicideNatDisasters NA              1
## 16 TurningIn     FALSE           1
## 17 TurningIn     TRUE           15
## 18 WorldsApart   FALSE           7
## 19 WorldsApart   TRUE          456
## 20 WorldsApart   NA              2

```